

Lecture 21 Surface Integrals Cont'd

Recap: Surface S parametrized by

$$\vec{r}(u, v) = x(u, v) \vec{i} + y(u, v) \vec{j} + z(u, v) \vec{k} \quad a \leq u \leq b \quad c \leq v \leq d$$

Scalar-valued function $f(x, y, z)$

$$\text{Surface integral } \iint_S f \, dS = \int_{v=c}^{v=d} \int_{u=a}^{u=b} f(\vec{r}(u, v)) \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| \, du \, dv$$

Surface S represented by $z = g(x, y) \quad (x, y) \in R$

$$\iint_S f \, dS = \iint_R f(x, y, g(x, y)) \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \, dx \, dy$$

Example. Let S be the sphere of radius a centered at the origin.

Evaluate the surface integral $\iint_S (x^2 + y^2) \, dS$.

$$\text{Sol. } \vec{r}(\theta, \phi) = (a \sin \phi \cos \theta) \vec{i} + (a \sin \phi \sin \theta) \vec{j} + (a \cos \phi) \vec{k} \\ 0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \pi$$

$$\frac{\partial \vec{r}}{\partial \theta} = (-a \sin \phi \sin \theta) \vec{i} + a \sin \phi \cos \theta \vec{j} + 0 \vec{k}$$

$$\frac{\partial \vec{r}}{\partial \phi} = a \cos \phi \cos \theta \vec{i} + a \cos \phi \sin \theta \vec{j} - a \sin \phi \vec{k}$$

$$\frac{\partial \vec{r}}{\partial \theta} \times \frac{\partial \vec{r}}{\partial \phi} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -a \sin \phi \sin \theta & a \sin \phi \cos \theta & 0 \\ a \cos \phi \cos \theta & a \cos \phi \sin \theta & -a \sin \phi \end{vmatrix}$$

$$= (-a^2 \sin^4 \phi \cos \theta) \vec{i} + (-a^2 \sin^4 \phi \sin \theta) \vec{j} + (-a^2 \sin \phi \cos^2 \phi) \vec{k}$$

$$\left| \frac{\partial \vec{r}}{\partial \theta} \times \frac{\partial \vec{r}}{\partial \phi} \right| = \sqrt{a^4 \sin^4 \phi + a^4 \sin^4 \phi \cos^2 \phi} = a^2 \sin \phi$$

$$x^2 + y^2 = a^2 \sin^2 \phi$$

$$\begin{aligned}
 \iint_S (x^2 + y^2) ds &= \int_0^{2\pi} \int_0^{2\pi} a^4 \sin^4 \phi \, a^4 \sin \phi \, d\theta \, d\phi \\
 &= a^4 \int_0^{2\pi} d\theta \int_0^{2\pi} \sin^5 \phi \, d\phi \\
 &= 2\pi a^4 \int_0^{2\pi} \sin^5 \phi \, d\phi \\
 &= \frac{8\pi a^4}{3}
 \end{aligned}$$

Example. Let S be the plane $3x + 2y + z = 1$ defined over the region $0 \leq x \leq 1$ and $0 \leq y \leq 1$. Compute the surface integral: $\iint_S (x + y + z) ds$.

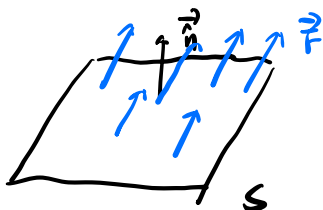
Sol. $z = 1 - 3x - 2y$

$$\frac{\partial z}{\partial x} = -3 \quad \frac{\partial z}{\partial y} = -2$$

$$\begin{aligned}
 \iint_S (x + y + z) ds &= \int_0^1 \int_0^1 (1 - 2x - y) \sqrt{(-3)^2 + (-2)^2 + 1} \, dx \, dy \\
 &= \sqrt{14} \int_0^1 \int_0^1 (1 - 2x - y) \, dx \, dy \\
 &= \sqrt{14} \int_0^1 [x - x^2 - xy] \Big|_{x=0}^{x=1} dy \\
 &= \sqrt{14} \int_0^1 -y \, dy \\
 &= -\frac{\sqrt{14}}{2}
 \end{aligned}$$

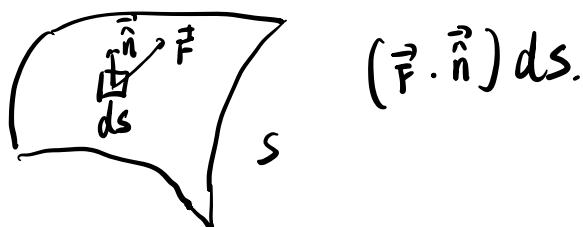
Surface flux.

Consider a constant vector field $\vec{F}$ and a flat plane



$(\vec{F} \cdot \vec{n})$ (area of the plane)

Consider a non-constant vector field $\vec{F}$ and a curved surface S

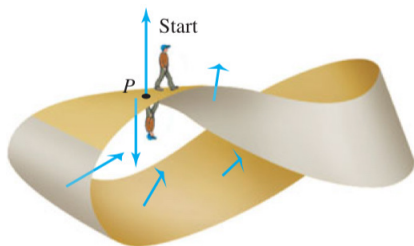


Def (Surface Flux). Given a vector field $\vec{F}$ and a surface S , the surface flux of $\vec{F}$ through S is defined to be the surface integral $\iint_S \vec{F} \cdot \vec{n} ds$.

where $\vec{n}$ denotes unit normal vector to S at each point.

Remarks: 1. For a closed surface such as a sphere, it is a convention to choose the outward unit normal.

2. We do not define surface flux on non-orientable surfaces.



Surface S : $\vec{r}(u, v)$ $a \leq u \leq b$, $c \leq v \leq d$

$$\vec{n} = \pm \frac{\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v}}{\left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right|} \quad dS = \left| \frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right| du dv$$

$$\begin{aligned} \iint_S \vec{F} \cdot \vec{n} ds &= \pm \iint_S \vec{F} \cdot \left(\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right) du dv \\ &= \pm \int_c^d \int_a^b \vec{F} \cdot \left(\frac{\partial \vec{r}}{\partial u} \times \frac{\partial \vec{r}}{\partial v} \right) du dv \end{aligned}$$

S : $z = g(x, y)$ $(x, y) \in R$.

$$\vec{r}(x, y) = \langle x, y, g(x, y) \rangle \quad \vec{F} = \langle F_x, F_y, F_z \rangle$$

$$\frac{\partial \vec{r}}{\partial x} = \langle 1, 0, \frac{\partial z}{\partial x} \rangle \quad \frac{\partial \vec{r}}{\partial y} = \langle 0, 1, \frac{\partial z}{\partial y} \rangle$$

$$\frac{\partial \vec{r}}{\partial x} \times \frac{\partial \vec{r}}{\partial y} = \langle -\frac{\partial z}{\partial x}, -\frac{\partial z}{\partial y}, 1 \rangle$$

$$\iint_S \vec{F} \cdot \vec{n} \, ds = \pm \iint_R (-F_x \frac{\partial z}{\partial x} - F_y \frac{\partial z}{\partial y} + F_z) \, dx \, dy.$$

Example. Consider $\vec{F} = -\mu m m \frac{x\vec{i} + y\vec{j} + z\vec{k}}{(x^2 + y^2 + z^2)^{3/2}}$.

Let S be part of the horizontal plane $z=1$ over the region $x^2 + y^2 \leq 1$.

Compute the surface flux $\iint_S \vec{F} \cdot \vec{n} \, ds$.

Here $\vec{n}$ is chosen to be the upward normal.

Sol. $\vec{r}(r, \theta) = (r \cos \theta)\vec{i} + (r \sin \theta)\vec{j} + \vec{k} \quad 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi$

$$\frac{\partial \vec{r}}{\partial r} = \cos \theta \vec{i} + \sin \theta \vec{j}$$

$$\frac{\partial \vec{r}}{\partial \theta} = -r \sin \theta \vec{i} + r \cos \theta \vec{j}$$

$$\frac{\partial \vec{r}}{\partial r} \times \frac{\partial \vec{r}}{\partial \theta} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos \theta & \sin \theta & 0 \\ -r \sin \theta & r \cos \theta & 0 \end{vmatrix} = r \vec{k}.$$

$$\vec{F} = -\mu m m \frac{(r \cos \theta)\vec{i} + (r \sin \theta)\vec{j} + \vec{k}}{(r^2 \cos^2 \theta + r^2 \sin^2 \theta + 1)^{3/2}}$$

$$= -\mu m m \frac{r \cos \theta \vec{i} + r \sin \theta \vec{j} + \vec{k}}{(r^2 + 1)^{3/2}}$$

$$\vec{F} \cdot \left(\frac{\partial \vec{r}}{\partial r} \times \frac{\partial \vec{r}}{\partial \theta} \right) = -\frac{\mu m m r}{(r^2 + 1)^{3/2}}$$

$$\iint_S \vec{F} \cdot \vec{n} \, ds = \int_0^{2\pi} \int_0^1 -\frac{\mu m m r}{(r^2 + 1)^{3/2}} \, dr \, d\theta.$$

$$= 2\pi \mu m m \int_0^1 -\frac{r}{(r^2 + 1)^{3/2}} \, dr.$$

$$= 2\pi \mu m m \left[(r^2 + 1)^{-1/2} \right]_{r=0}^{r=1} = 2\pi \mu m m \left(\frac{1}{\sqrt{2}} - 1 \right)$$

Example. Let $\vec{F} = x\vec{i} - y\vec{j}$. Find the upward flux of $\vec{F}$ over S which is the upper part of the sphere with radius $\sqrt{2}$ centered at the origin by the plane $z=1$.

Sol.



$$\vec{r}(\theta, \phi) = \langle \sqrt{2} \sin \phi \cos \theta, \sqrt{2} \sin \phi \sin \theta, \sqrt{2} \cos \phi \rangle$$

$$0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \frac{\pi}{4}.$$

$$\frac{\partial \vec{r}}{\partial \theta} = (-a \sin \phi \sin \theta) \vec{i} + a \sin \phi \cos \theta \vec{j} + 0 \vec{k}$$

$$\frac{\partial \vec{r}}{\partial \phi} = a \cos \phi \cos \theta \vec{i} + a \cos \phi \sin \theta \vec{j} - a \sin \phi \vec{k}.$$

$$\frac{\partial \vec{r}}{\partial \theta} \times \frac{\partial \vec{r}}{\partial \phi} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -a \sin \phi \sin \theta & a \sin \phi \cos \theta & 0 \\ a \cos \phi \cos \theta & a \cos \phi \sin \theta & -a \sin \phi \end{vmatrix} \quad a = \sqrt{2}$$

$$= (-a^2 \sin^2 \phi \cos \theta) \vec{i} + (-a^2 \sin^2 \phi \sin \theta) \vec{j} + (-a^2 \sin \phi \cos \phi) \vec{k}.$$

$$= -2 \sin^2 \phi \cos \theta \vec{i} - 2 \sin^2 \phi \sin \theta \vec{j} - 2 \sin \phi \cos \phi \vec{k}.$$

$$\vec{F} = x\vec{i} - y\vec{j} = \sqrt{2} \sin \phi \cos \theta \vec{i} - \sqrt{2} \sin \phi \sin \theta \vec{j}$$

$$\vec{F} \cdot \left(\frac{\partial \vec{r}}{\partial \theta} \times \frac{\partial \vec{r}}{\partial \phi} \right) = -2\sqrt{2} \sin^3 \phi (\cos^2 \theta - \sin^2 \theta)$$

$$= -2\sqrt{2} \sin^3 \phi \cos 2\theta.$$

$$\iint_S \vec{F} \cdot \vec{n} \, dS = \int_0^{2\pi} \int_0^{\pi/4} \vec{F} \cdot \left(-\frac{\partial \vec{r}}{\partial \theta} \times \frac{\partial \vec{r}}{\partial \phi} \right) d\phi d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi/4} 2\sqrt{2} \cos 2\theta \sin^3 \phi \, d\phi d\theta.$$

$$= 2\sqrt{2} \left(\underbrace{\int_0^{2\pi} \cos 2\theta d\theta}_0 \right) \left(\int_0^{\pi/4} \sin^3 \phi d\phi \right) = 0$$

Physical interpretations of surface flux

Example. $\vec{u}$ is the velocity vector field of fluid (in ms^{-1})

The surface area of S has unit m^2 .

$\iint_S \vec{u} \cdot \vec{n} \, dS$ has unit $\text{m}^3 \text{s}^{-1}$, it measures the net volume of fluid through the surface S in the direction of $\vec{n}$.

Example. $\vec{E}$ is a electric field and S is a closed surface.

$$\oiint \vec{E} \cdot \vec{n} \, dS = \frac{1}{\epsilon_0} \iiint \rho \, dV. \quad \text{by Gauss's Law}$$

ρ : charge density

Example. $\vec{B}$ is a magnetic field and S is a closed surface

$$\oiint \vec{B} \cdot \vec{n} \, dS = 0 \quad \text{by Gauss's Law for Magnetism.}$$

Maxwell's equations (2 of 4)

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \cdot \vec{B} = 0.$$